

3a	Specific latent heat of vaporization is the <u>amount of thermal energy per unit mass</u> required to change a substance from <u>liquid to vapour</u> <u>without change of temperature</u> .	B1 B1
3b(i)	Increase in potential energies of all the water molecules when the water boils.	B1
3b(ii)	$\Delta U = Q + W$ $Q = mL$ (positive because heat supplied) = $1000 \times 2300 \text{ J}$ $W = p\Delta V$ (negative because expansion) $= p (m / \rho) \times 10^3 = - 1.0 \times 10^5 \times (1.0 / 1000) \times 10^3 \text{ J}$ $\Delta U = 1000 \times 2300 - 1.0 \times 10^5 \times (1.0 / 1000) \times 10^3 \text{ J}$ $= 2.2 \times 10^6 \text{ J}$	C1 C1 C1 A1
3b(iii)	Part of the heat supplied is used to do work against the atmosphere when water expands from liquid to vapour.	B1

4a	From Fig. 4.1, wavelength $\lambda = 0.20 \text{ m}$	C1
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